D03 · Industrial Automation & Robotics

05 · Automation strategy & ROI methodology

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Automation strategy and ROI methodology is the structured discipline of evaluating which automation investments to make, in what sequence, with what business case. The methodology spans (1) strategic fit (does this automation advance the operations strategy), (2) economic case (multi-dimensional ROI: labor, throughput, quality, opportunity cost, risk reduction, strategic optionality), (3) sequencing (which automation precedes which, and why), and (4) sustainability (organizational readiness to absorb the change). Operations strategies that lack this discipline either over-invest in the wrong cases or under-invest in the right ones — usually both.

2 WHY IT MATTERS

Automation ROI methodology is one of the most consequential and most consistently mis-practiced disciplines in operations. **The dominant failure mode is the labor-savings-only business case — automation proposals justified solely on headcount reduction, with no accounting for cycle time gain, quality improvement, capacity flexibility, opportunity cost of not automating, or strategic optionality. The methodology rejects the right projects and approves the wrong ones — usually both simultaneously**. The peer-reviewed text on operations strategy and technology investment treats this gap directly **(Hayes, Pisano, Upton & Wheelwright, 2005 — peer-reviewed, "Operations, Strategy, and Technology", Wiley)**: technology investments must be evaluated as multi-dimensional strategic decisions, not as single-variable financial returns. Industry data confirms the asymmetry — **automation projects evaluated on labor savings alone show acceptable ROI in only 20–30% of cases; the same projects evaluated with multi-dimensional ROI (labor + cycle time + quality + opportunity cost) show acceptable ROI in 55–70% of cases, with the additional approvals concentrated in the strategically important projects that labor-only analysis killed**. The methodology gap is not analytical; it is what gets counted.

3 HOW IT IS DONE

1. **Build the automation portfolio against the operations strategy — not against this year's capex budget.** The portfolio question is which automation cases the operation needs to be capable of in 3–5 years (variable-mix high-mix capability, lights-out night-shift operation, regulatory traceability, quality at single-digit ppm), and what sequence of investments builds that capability. Plants that build their automation portfolio one capex cycle at a time end up with the automation that fit each year's budget; plants that build against a 5-year capability roadmap end up with the automation the business needed.
2. **Score every automation case on six dimensions, not on one — labor, throughput, quality, opportunity cost, risk reduction, strategic optionality.** Labor savings: direct headcount and labor hour reduction at fully-loaded rate. Throughput gain: cycle time reduction and capacity increase, monetized at gross margin per unit. Quality gain: defect rate reduction and COQ impact. Opportunity cost: revenue lost or competitive position eroded by not automating. Risk reduction: ergonomic injury risk, regulatory non-compliance risk, single-point-of-failure dependency risk. Strategic optionality: capabilities the investment enables for future decisions. A case that fails labor-only but passes the six-dimensional view is a strategically important project the labor-only methodology would have killed.
3. **Quantify the opportunity cost of not automating — it is usually the largest term and the most commonly omitted.** The opportunity cost is what happens to the operation, the customer relationship, and the competitive position over the time period of the investment if automation is not deployed. Examples: customer-mandated traceability requirement triggering account loss at end of contract; competitor automation move that captures a price-point segment; regulatory change requiring per-unit serialization that manual operation cannot meet. Plants that omit opportunity cost calculate the cost of doing something and ignore the cost of not doing it; the latter is frequently larger and almost always missing from the business case.
4. **Use NPV with explicit assumptions over IRR or simple payback — the methodology choice constrains the case structure.** Simple payback ignores time value and back-loaded benefits; IRR is unstable for projects with mixed cash flows; NPV with documented assumptions (discount rate, time horizon, terminal value, sensitivity ranges) forces the disciplined economic conversation. The methodology choice is not academic; payback-based cases consistently under-weight strategic optionality, IRR-based cases consistently mis-rank projects with different scale, and NPV-based cases force the assumption

transparency that survives audit.

5. **Sequence the portfolio around strategic dependencies — what automation enables what other automation.** Automation projects compound: an MES deployment enables data-driven SPC deployment, which enables in-process vision control, which enables real-time OEE management. Plants that sequence the portfolio randomly (whichever case has the easiest business case this quarter) end up with high-cost integrations between disconnected automation islands. Plants that sequence around dependencies pay slightly more for early investments and capture compounding leverage over the multi-year program.
6. **Govern the portfolio quarterly with operations + finance + engineering at the table — not project-by-project with finance as gatekeeper.** The most common organizational pathology is automation cases evaluated one-by-one by finance using a labor-savings template. The structural fix is a quarterly automation-portfolio review with operations leadership (strategic fit), engineering (technical feasibility and sequencing), and finance (economic case and capital allocation) at one table. Decisions about the portfolio sequence, the cases to fund, and the cases to defer get made jointly with multi-dimensional analysis; finance veto power on cases that fail labor-only metrics goes away. The single biggest determinant of automation portfolio quality is whether this review exists.

WORKED EXAMPLE — ILLUSTRATIVE

A medical-device manufacturer (FDA-regulated, 3 product families, $180M annual revenue) submits an automation proposal for a sterile-packaging line to capital planning for the third consecutive year. The labor-savings business case: **$640K capex, $152K/year labor savings (3.2 FTEs at fully-loaded $47.5K), simple payback = 4.2 years, NPV at 10% discount over 7 years = -$48K**. Capital planning rejects on the labor-only ROI for the third time. The fourth submission re-frames the case with six-dimensional analysis: **(1) labor savings $152K/year (unchanged); (2) throughput gain — cycle time reduction 23% releases capacity worth $340K/year in additional product sales at current gross margin; (3) quality gain — eliminates manual handling that drives 0.4% non-sterile rejection, worth $180K/year in avoided rework and customer credits; (4) opportunity cost — major customer signaled in 2023 audit that manual sterile packaging is no longer acceptable; failure to automate puts $14M annual revenue at risk over the next 3-year contract cycle; (5) risk reduction — repetitive-strain ergonomic claims on this operation averaging $67K/year; (6) strategic optionality — automated line is prerequisite for serialization compliance under upcoming regulatory change**. Revised case: **NPV at 10% over 7 years = +$1.42M, simple payback 1.6 years, opportunity-cost-adjusted breakeven under 1 year**. Same project, same hardware, same vendor — different methodology. Approved at the next portfolio review. The 4 previous rejections were not analytical errors; they were methodological errors that the labor-only framework was structurally incapable of correcting.

IN PRACTICE

Score every case on six dimensions, not one

Labor + cycle time + quality + opportunity cost + risk reduction + strategic optionality **(Hayes, Pisano, Upton & Wheelwright, 2005 — peer-reviewed, Wiley)**. Automation projects evaluated on labor alone show acceptable ROI in 20–30% of cases; the same projects with multi-dimensional ROI show acceptable ROI in 55–70%.

Quantify opportunity cost as carefully as labor savings

The cost of not automating is hidden — lost market share when a competitor automates first, retained inefficiency that compounds, capacity that does not scale with demand growth. The labor-savings-only template makes this invisible by construction; the multi-dimensional template forces it onto the page.

Build the portfolio against the operations strategy, not this year's capex budget

Capex availability drives sequencing; strategy drives composition. Plants that invert this — letting budget cycles drive which automation gets approved — accumulate a portfolio that is financially convenient and strategically marginal. The CFO and COO must agree on the strategy before the budget gets cut.

Joint methodology: operations + finance + engineering

Finance owns capex templates; operations submits cases against those templates; the aggregate is a methodology that underweights what operations cares about. The structural fix is co-ownership: methodology jointly owned, dimensions jointly scored, governance jointly chaired.

WHAT YOU WOULD USE

AUTOMATION STRATEGY & PORTFOLIO

Anaplan / Oracle Hyperion / SAP IBP (capex and portfolio planning modules) — Enterprise capital-portfolio platforms that integrate operations capex with financial planning, scenario analysis, and multi-dimensional ROI tracking; the right fit for multi-plant operations with $50M+ annual capex programs **(Anaplan — vendor; Oracle — vendor; SAP — vendor)**.

Sciforma / Planview / Smartsheet for project portfolio management — Project-portfolio platforms that span engineering and finance for capex governance — case templates, gate reviews, portfolio dashboards, dependency tracking; the layer between strategy and execution **(Planview — vendor; Smartsheet — vendor)**.

External operations strategy consulting + internal portfolio governance team — For operations building the methodology — strategy houses (BCG, McKinsey, Bain, Deloitte) for the initial methodology design, internal portfolio team (typically a finance lead + operations strategy lead + engineering analyst) to sustain it. The most common path to a working methodology where one did not previously exist **(Deloitte — vendor)**.

A jointly owned Excel template + quarterly portfolio review + finance-operations co-ownership — For operations without platform investment — a multi-dimensional case template with NPV calculation, sensitivity analysis, and opportunity-cost section; jointly owned by CFO and COO; reviewed quarterly with operations + finance + engineering at the table. The lowest-tech option produces the best outcome when the methodology and the governance are real; the template is the artifact, the joint ownership is the system.

4 SO WHAT

Automation justified on labor savings alone rejects the right projects and approves the wrong ones — the single-dimensional business case takes every capex decision incorrectly. Build the portfolio against the operations strategy not against this year's capex budget, score every case on six dimensions not one, quantify opportunity cost as carefully as labor savings, use NPV with explicit assumptions over payback or IRR, sequence the portfolio around strategic dependencies, and govern quarterly with operations + finance + engineering jointly. The methodology is the bottleneck — the same automation projects that fail single-dimensional analysis pass multi-dimensional analysis, and the difference between operations that build the automation capability they need and operations that buy the automation they could justify lives entirely in which methodology is on the wall.

From the SCRA v2.2 reference architecture · mrsupplychain.ai · © 2026 Ismael López · Mr. Supply Chain® México